



Vansh Bhandari

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B.Tech - Chemical Science and Technology

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	7.53 (Current)	2023-Present
Senior Secondary	CBSE Board	94.2%	2023
Secondary	CBSE Board	96.6%	2021

EXPERIENCE

- Quantitative Research Consultant** Remote
WorldQuant BRAIN Jan 2025 – Present
 - Utilized a **quantitative approach** to develop, backtest, and optimize **alpha trading strategies** using historical market data.
 - Engineered **60+ alpha strategies** with **Sharpe ratios exceeding 1.25** and **low pairwise correlation (< 0.7)** across **multiple asset classes** and **geographies** including the **USA, Europe, and Asia**, focusing on performance consistency and diversification.

PROJECTS

- Trading Strategy for BTC/USDT and ETH/USDT Market** Oct. 2024 – Dec. 2024
Inter IIT Tech Meet 13.0, IIT Bombay GitHub
 - Designed **trading strategies** using **Reinforcement Learning** for **BTC/USDT** and **correlation analysis** for **ETH/USDT**.
 - Applied techniques like **CUSUM analysis** and **Q-learning** to enhance decision-making and strategy robustness.
 - Achieved **224% profit** on BTC/USDT market (**42.68% over benchmark**) and **7684% profit** on ETH/USDT (**355.48% over benchmark**), with respective win rates of **68%** and **48%**, showcasing strong performance and consistency in volatile markets.
- Pairs Trading Project** Nov. 2024 – Dec. 2024
Finance and Economics Club, IIT Guwahati GitHub
 - Developed a **mean-reversion pairs trading strategy** using **cointegration analysis** on **Eicher Motors** and **Motherson**, generating **z-score signals** and backtesting on **NSE equity data** with **Sharpe Ratio** and **drawdown** metrics.
 - Executed **OLS regression** and **p-value testing** to confirm cointegration; computed **z-scores** to identify trade Entry/Exit points.
 - Achieved a **54% cumulative return** from 2017 to 2020 through backtesting, significantly outperforming the market.
- Approximation of Option Greeks Using Artificial Neural Networks** Jan. 2025 – Mar. 2025
Finance and Economics Club, IIT Guwahati GitHub
 - Built an **ANN-based model** to estimate **Option Greeks** under **Piterbarg Model**, enhancing speed in pricing and risk analysis.
 - Predicted third month's **Delta** by training model on 2-months Option data with high precision (**R^2 : 0.998, MAE: 0.008**).
 - Enabled faster, scalable, and real-time **derivatives risk assessment** through ANN-based approximation.
- Banking Loan Default Prediction** May. 2025 – Jun. 2025
State Bank Of India – Finnovation'25 GitHub
 - Built a **classification model** using **100+ financial and behavioral variables** to identify non-defaulters in loan applicants.
 - Performed **data preprocessing**, **feature engineering**, **class balancing**, and **SHAP-based selection**; then trained and evaluated **XGBoost** and **LightGBM** models achieving **91.3% accuracy**, **F1-score of 0.60**, and **precision of 0.60** on the test set.
 - Deployed a **prototype interface** enabling loan officers to receive instant default predictions for informed decision-making.

TECHNICAL SKILLS

- Programming:** C/C++, Python, Pine Script
- Finance:** Fundamental Analysis, Technical Analysis, Financial Modelling
- Data Visualization:** MS Excel, SQL, Power BI, Tableau, pandas, Matplotlib, Seaborn

KEY COURSES TAKEN

- Mathematics:** Linear Algebra, Complex Analysis, Real Analysis
- Machine Learning:** Machine Learning Specialisation Course
- Finance and Economics:** Fundamental Analysis Bootcamp, Introductory Options, Econometrics, Finlatics Financial Analyst

POSITIONS OF RESPONSIBILITY

- Overall Coordinator**, Finance and Economics Club, IIT Guwahati Mar. 2025 – Present
 - Leading a team of **100+ sophomores**, mentoring them in **quant**, **capital markets**, and **DeFi**.
 - Conducted **3+ finance courses**, including **2 PAN India-level programs**, along with **5+ events** and **15+ webinars**.
 - Co-authored the **Union Budget 2024–25 Analysis Report**, analyzing key **fiscal and policy** implications.

ACHIEVEMENTS

- Gold Medal**, Inter IIT Tech Meet 13.0, Bombay, among all IIT's in Cryptocurrency Strategy PS 2024
- Finalist**, Finnovation – SBI PSB National Hackathon, In Top 10 teams out of 200+ participated teams 2025
- JEE Advanced 2023**, Secured **AIR 7481** among 0.18 million candidates 2023
- JEE Mains 2023**, Secured **AIR 9042** among 1.1 million candidates 2023
- NDA Written Qualified**, Cleared UPSC NDA Exam (Written Round) 2022

EXTRACURRICULARS

- Senior Executive**, Entrepreneurship Cell, IIT Guwahati, Led startup and innovation initiatives. 2024
- Public Relations & Branding Manager**, UDGAM, for North-East India's largest entrepreneurship summit 2025
- Organizer**, Vittharth – National Finance Competition, Conducted a PAN-India finance competition 2025
- Silver Medalist**, Dance Manthan, Secured 2nd place in IIT Guwahati's Inter-Hostel Dance Competition 2024
- Bronze Medalist**, Yoga – Spardha (Inter- hostel Sports Meet), Secured 3 Bronze medals 2025